

Professional Ethics & Research Misconduct

What does “professional ethics” in science/research mean?

- Being a “good person” – using that as a motivator for not manipulating science/facts/people
- Manipulate → perpetuate something you know to be untrue, for your own benefit
- Not citing appropriately/taking credit for things you did not do.
- Taking advantage of power dynamics
- Being able to collaborate constructively → Promoting/maintaining Healthy work environments
- Prioritizing human health and safety over scientific results
- Following mutually agreed upon standards – implicit/explicit guidelines
???? → e.g. establishing in advance a code of conduct
- Keeping TRUST – with the public/peers/community/scientists
- Reporting results in a way that can be externally validated/reproduced

What does “professional ethics” in science/research mean?

Clear citations – giving appropriate credit, attributing work to the correct person

Giving credit to junior folks

Not falsifying data, accurately presenting your results

Appropriate behavior – with facilities and with colleagues

Don't abuse your position as an expert (media, public facing).

Harassment, abuse, discrimination

What does “professional ethics” in science/research mean?

APS task force on ethics

(Kate Kirby & Frances A. Houle, 2004):

- Truthful, careful handling and reporting of data
- Responsible, respectful interactions with colleagues and subordinates
- Adherence to publication guidelines, including proper recognition of research contributions

Research misconduct is defined under *federal policy* as **fabrication**, **falsification** or **plagiarism** in proposing, performing or reviewing research, or in reporting research results.

Research misconduct does not include honest errors or differences of opinion.

Research Misconduct

- **Fabrication**: making up data or results and recording or reporting them
- **Falsification**: manipulating research materials, equipment, or processes, or changing or omitting data or results such that the research is not accurately represented in the research control
- **Plagiarism**: the appropriation of another person's ideas, processes, results or words without giving appropriate credit

Research Misconduct

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-  Data falsification
 -  Not including appropriate authors
 -  Plagiarism
 -  Less than truthful reports
 -  Not citing prior work
 -  Delaying referee reports
 -  Including inappropriate authors
 -  Other

Distribution of ethics violations categories of which junior members of the APS reported having personal knowledge. Of the junior members responding to the APS ethics survey 39% cited one or more of these transgressions

Fabrication of Data

(Ethics in Engineering Practice and Research by Caroline Whitbeck)

- 1998: Victor Ninov claimed the discovery of two new elements (“super-heavy” element called 118 and decay product 116)
- Ninov’s colleagues and coauthors left it to him alone to deal with the raw data – only he knew how to run the computer programs that analyzed the data
- Other investigators were not able to replicate the experimental results
- Officials at Lawrence Berkeley National Lab investigated in 2002
- Computer log file found with evidence that data had been cut and pasted and numeric values were changed
- Ninov was fired, **coauthors reprimanded** and news release made to withdraw the discovery

Example

A CAREER IN THE BALANCE

Francine was just months away from finishing her Ph.D. dissertation when she realized that something was seriously amiss with the work of a fellow graduate student, Sylvia. Francine was convinced that Sylvia was not actually making the measurements she claimed to be making. They shared the same lab, but Sylvia rarely seemed to be there. Sometimes Francine saw research materials thrown away unopened. The results Sylvia was turning in to their common thesis advisor seemed too clean to be real.

Francine knew that she would soon need to ask her thesis advisor for a letter of recommendation for faculty and postdoc positions. If she raised the issue with her advisor now, she was sure that it would affect the letter of recommendation. Sylvia was a favorite of her advisor, who had often helped Sylvia before when her project ran into problems. Yet Francine also knew that if she waited to raise the issue the question would inevitably arise as to when she first suspected problems.

Both Francine and her thesis advisor were using Sylvia's results in their own research. If Sylvia's results were inaccurate, they both needed to know as soon as possible.

1. Should Francine first try to talk with Sylvia, with her thesis advisor, or with someone else entirely?
2. Does she know enough to be able to raise concerns?
3. Where else can Francine go for information that could help her decide what to do?

On Being a Scientist: Responsible Conduct in Research 2000, The National Academy of Sciences

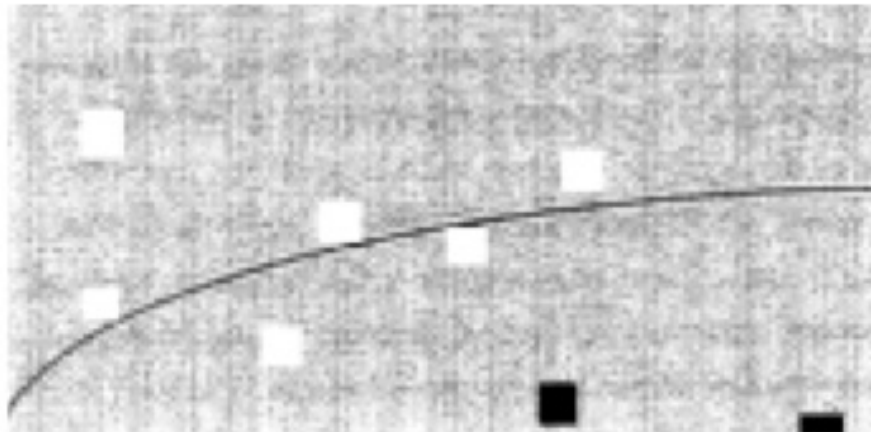
Go to : Grad Director, Dep Department Head

UA Ombuds Office : <https://ombuds.arizona.edu/>

Falsification of Data: Expected Results

THE SELECTION OF DATA

Deborah, a third-year graduate student, and Kathleen, a post-doc, have made a series of measurements on a new experimental semiconductor material using an expensive neutron source at a national laboratory. When they get back to their own laboratory and examine the data, they get the following data points. A newly proposed theory predicts results indicated by the curve.



During the measurements at the national laboratory, Deborah and Kathleen observed that there were power fluctuations they could not control or predict. Furthermore, they discussed their work with another group doing similar experiments, and they knew that the other group had gotten results confirming the theoretical prediction and was writing a manuscript describing their results.

In writing up their own results for publication, Kathleen suggests dropping the two anomalous data points near the abscissa (the solid squares) from the published graph and from a statistical analysis. She proposes that the existence of the data points be mentioned in the paper as possibly due to power fluctuations and being outside the expected standard deviation calculated from the remaining data points. "These two runs," she argues to Deborah, "were obviously wrong".

Confirmation Bias

- **Confirmation bias** is the tendency to search for, interpret, favor, and recall information in a way that confirms one's preexisting beliefs or hypotheses - Wikipedia
- Proposals set us up for this.

APS task force on ethics: what is the origin of research misconduct?

“By far the highest response rate and the most extensive and heartfelt answers .. came from the junior members of APS”

- Tenure review boards need to stop rewarding the Science/Nature culture above all else
- Our scientific community promotes the search of the surface and superficiality [to the] detriment of content and deepness
- Many breaches of ethics arise from **the pressure to publish** ...
- Belief that the researcher .. will be judged [by] the number of articles .. appearing on the CV. He or she will not be judged [by] the work spent on each paper, how many backup checks were performed to confirm the results and so on. ... for many people it **is more important to publish spectacular results than to publish true results.**

Drivers of Fabrication/Falsification

- APS Report: the emergence, over the past 15 years, of a "research system [that] stimulates continuously the competition in fashionable subjects in search of spectacular results,"
- "there is enormous pressure to do quality work in a short period of time" that is difficult or impossible to live up to.
- Some physicists may not be equipped to handle the pressure if they feel that their careers are at stake.

Treatment of subordinates

Abuse of graduate students by advisers:

- Threatening not to write letter of recommendation
- Students not getting credit for research
- Unreasonable expectations (work hours)

“Particularly shocking to the task force was how often the words ‘abuse’ and ‘**exploitation**’ were used to describe the treatment of graduate students”.

APS Statement 04.1

Subordinates should be treated with respect and with concern for their well-being. Supervisors have the responsibility to .. provide a safe, supportive working environment and fair compensation, and to promote the timely advance of graduate students and young researchers to the next stage of career development. In addition, supervisors should ensure that subordinates know how to appeal decisions without fear of retribution.

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What is missing?

- Coersion/Power Dynamics
- Nothing about the relations between PEOPLE
- Treatment of subordinates/students → sexual harassment, racial discrimination, bias, bullying --> dual anonymous
- Environment → conducive to equitable participation.
- Mental illness → hostile environments

What is missing?

- Sexual Harassment
- Racial Discrimination
- Bias
- Treatment of Subordinates – value of mentorship
- Land Use & Relationships with Indigenous Communities
- Equity of access to resources: facilities, education, training (leadership/technical)
- Accountability??
- Assessment ? (Data Collection) ?

Decadal Surveys:

Astronomy 2020, Planetary Science 2023

- Every 10 years (white papers year before)
- Six astronomy and astrophysics decadal surveys have been produced by the National Academies of Sciences, Engineering, and Medicine to date.
- While each discussed the state of the profession and societal impacts, recent ones are the first to feature a formal panel devoted to these concerns. 2020 panel is the first to feature the findings in the primary report.
<https://www.nationalacademies.org/read/26141/chapter/24#502>
- Planetary Science 2023 included DEIA topics for the first time in the history of its decadal.
<https://www.nationalacademies.org/read/26522/chapter/18#417>

The promise of the decadal survey is our collective power to influence our future



Panel on the State of the Profession and Societal Impacts

Appendix N

CHARGE:

The Panel on State of **the Profession** and Societal Impacts will gather information on the health and demographics of the astronomy and astrophysics community and **make actionable suggestions** to the Astro2020 committee on the topics of demographics, diversity and inclusion, workplace climate, workforce development, education, public outreach, and relevant areas of astronomy and public policy. The panel's suggestions will be incorporated into a program for all of astronomy and astrophysics by the Astro2020 committee.

Culture

- All scientific disciplines are knowledge communities, with ways of knowing, distinctive language, and beliefs about what types of questions are most important to pursue.
- The knowledge that communities produce, including scientific communities, are therefore cultural knowledge
- Stated another way, “scientific knowledge is but a particular form of cultural knowledge.” – Lauren Chambers

<https://baas.aas.org/pub/2020n7i162/release/1>

Culture & Values

- Resource prioritization is set by implicit values
- While implicit cultural values can be learned over time, “implicit” values favor those already living within these norms and thus embeds structural inequity.
- By leaving ***implicit*** the values that have guided past decadal surveys’ scientific priorities, the Profession failed to move forward together toward a more equitable future

Motivation for Astronomy & Astrophysics Values

- Decadal survey itself is “an intentional examination and enactment of the culture of astronomy”
- Current values are **implicit** and subject to interpretation → inequity thrives
- **Explicit**, shared community values reduces ambiguity and provides avenues for accountability
- Allows for collective sensemaking* around equitable science practices (*process by which people give meaning to collective experiences)
- **Proposal**: Shared Values that set Priorities, where Priorities are used to Assess Merit